4th Semester		
ISCED Code	**Course Code**	**Course Title**
0613	CSE-2424	Database Management System Lab
Credit Hours: 3	**Contact Hours:** 3	**Type:** Core, Engineering
Prerequisite: CSE-2423 (Database Management System)		
Co-requisite: CSE-2423 (Database Management System)		

Instructor's details, Consultation Hour and Office Locations

Instructor: **Dr. Mohammad Aman Ullah**
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Contact: 01815641524
Office Location: Room # 405, Computer Science Academic Building
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Consultation Hours: Saturday (10:00 – 12:30), Sunday (08:00 – 8:30), Tuesday (8:00 – 8:30)

Course Assessments	**CIE**: Continuous Internal Evaluation	Attendance	10 Marks
		Assignment and Project	30 Marks
		Mid-term	20 Marks
	SEE: Semester End Examination		40 Marks
	Total		100 Marks

Course Rationale:

The Database Management Systems (DBMS) course is designed to equip students with a comprehensive understanding of data organization, storage, and retrieval techniques essential for modern applications. By focusing on SQL for relational database management, PL/SQL for advanced procedural programming within databases, and MongoDB for non-relational, document-based data handling, this course prepares students to design, query, and manage databases efficiently. The integration of these skills enables learners to address diverse real-world challenges in data management across various industries, fostering expertise in both traditional and contemporary database paradigms.

Objectives:

Upon completion of the course, students will be able to:

1. Familiarity with SQL, PL/SQL, NoSQL and fundamental of Database Management System
2. Ability to apply SQL, PL/SQL, and NoSQL for data searching and developing different types of database applications
3. Apply different concepts to Database management systems such as Integrity, Security, Normalization, Indexing, Transaction, Recovery system, Distributed system.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage (%)
1.	**Demonstrate** familiarity with SQL, PL/SQL, NoSQL and fundamental of Database Management System	50 %
2.	**Create** different types of database applications using SQL, PL/SQL, and NoSQL	30 %
3.	**Apply** different concepts for Database management such as Integrity, Security, Normalization, Indexing, Transaction, Recovery system, Distributed system.	20 %

Mapping of CLO-PLO-DL-KP-EP-EA:

#	CLOs	PLOs	DL	KP	EP	EA	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO1	**Demonstrate** familiarity with SQL, PL/SQL, NoSQL and fundamental of Database Management System	PLO1		2,3	-	-	Lecture, Class discussion, Assignment, Lab work,	Exam, Quiz, Assignment
CLO2	**Create** different types of database applications using SQL, PL/SQL, and NoSQL	PLO3, PLO9		K,	-	-	Lecture, Class discussion, Assignment, Lab work,	Assignment, Quiz, Exam
CLO3	**Apply** different concepts for Database management such as Integrity, Security, Normalization, Indexing, Transaction, Recovery system, Distributed system.	PLO2	C3	2,3,5	-	-	Lecture, Assignment, Quiz	CT, Assignment
Note: DL: Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities, **Learning Domains** (C: Cognitive, A: Affective, P: Psychomotor)								

Course Content:

#	Content	Duration	CLOs
1	Introduction to SQL, Relational Database Management System.	1 week	CLO1

	Oracle 12c: Object Relational Database Management System, SQL statements, about PL/SQL and its environments.		
2	Writing Basic SQL statements, Capabilities of SQL SELECT Statements, Restricting and sorting data.	1 week	CLO2
3	Single-Row-Functions	1 week	CLO2
4	Displaying Data from multiple tables	1 week	CLO2
5	Aggregating data using Group Functions. Sub queries, Multiple-Column Subqueries	2 weeks	CLO2
6	Manipulating Data, Creating and Managing Tables including constraints. Instant Database Creation on some unknown domain, Other Database Objects, Controlling User Access.	3 weeks	CLO3
7	PL/SQL, Declaring Variables, writing Executable Statements. Working with Composite Data types. Writing Explicit Cursors	2 weeks	CLO1, CLO2
8	Introduction to NoSQL Database, Introduction & Overview of MongoDB, MongoDB Installation, CRUD Operation in MongoDB	1 week	CLO1, CLO2

Weekly Activity Plan:

Week	Topic	TLS	AS	CLOs
Week 1	Topic: ● Introduction to SQL, Relational Database Management System. ● Oracle 12c: Object Relational Database Management System, SQL statements, about PL/SQL and its environments.	Lecture		CLO1
Week 2	● Solving problems from the textbook practice section. ● **Topic**: Writing Basic SQL statements, Capabilities of SQL SELECT Statements, Restricting and sorting data.	Lecture, Problem solving	Problem solving	CLO2
Week 3	● Solving problems from the textbook practice section. ● **Topic**: Single-Row-Functions	Lecture, Problem solving	Problem solving	CLO2
Week 4	● Solving problems from the textbook practice section. ● **Topic**: Displaying Data from multiple tables	Lecture, Problem solving	Problem solving	CLO2
Week	● Solving problems from the textbook	Lecture,	Problem	CLO2

Week	Topic	TLS	AS	CLOs
5	practice section. ● **Topic**: Aggregating data using Group Functions.	Problem solving	solving	
Week 6	Review of the midterm exam content			
Week 7	● Solving problems from the textbook practice section. ● **Topic**: Sub queries, Multiple-Column Subqueries	Lecture, Problem solving	Problem solving	CLO2
Week 8	● Solving problem from textbook practice section ● **Topic**: Manipulating Data, Creating and Managing Tables including constraints.	Lecture,Problem solving	Problem solving	CLO3
Week 9	● Instant Database Creation on some unknown domain	Lecture, Problem solving	Problem solving	CLO3
Week 10	● Solving problem from textbook practice section ● **Topic**: Other Database Objects, Controlling User Access.	Lecture, Problem solving	Problem solving	CLO3
Week 11	**Topic**: ● PL/SQL, Declaring Variables, writing Executable Statements. ● Working with Composite Data types.	Lecture, Problem solving	Problem solving	CLO1, CLO2
Week 12	● Solving problem from textbook practice section ● **Topic**: Writing Explicit Cursors	Lecture, Problem solving	Problem solving	CLO2
Week 13	Lab test 2			
Week 14	● Solving problem from textbook practice section ● **Topic**: Introduction to NoSQL Database, Introduction & Overview of MongoDB, MongoDB Installation, CRUD Operation in MongoDB	Lecture, Problem solving	Problem solving	CLO1, CLO2
Week 15	Review of the semester final content			

Marks Distribution:

Course Assessment Pattern (Lab courses):

		CIE (50 marks)		**SEE (50 marks)**
Cognitive learning	Affective	Attendance	Continuous	Final Exam /

	learning	Marks (10)	Evaluation (50)	Project (40)
Remember				
Understand				
Apply				
Analyze				
Evaluation				
Create				
	Responding	10		
Total allocated marks		10	50	40

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

1. Kock and Loney, Oracle 11g the Complete Reference, 11th ed., Oracle press/ McGraw-Hill, 2012, ISBN:978-0-07-159876-7
2. Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Database System Concept, 6th ed., McGraw-Hill, 2011, ISBN : 978-0-07-352332-3
3. RamezElmasri, Shamkant B. Navathe, Fundamentals of database systems, 6th ed., Pearson Education, 2011, ISBN : 10: 987-0-136-08620-9

Reference Books:

1. Hector Garcia-Molina, Jeff Ullman, and Jennifer Widom, Database Systems: The Complete Book, 3rd ed., Pearson (Addison-Wesley + Prentice-Hall) 2008, ISBN : 9870138613370
2. Connolly, Thomas M., Begg, Carolyn E. Database systems : a practical approach to design, implementation, and management, 4th ed., Addison-Wesley, 2005, ISBN : 987321210255